

Summer Math Adventure

W4
D1

Week 4 · Day 1 | Decimals: Place Value & Reading/Writing

Name: _____

Date: _____

DECIMALS

Decimal Place Value Chart:

Hundreds | Tens | Ones | Tenths | Hundredths | Thousandths

1 3 7 . 4 0 5

Read as: "one hundred thirty-seven AND four hundred five thousandths" (137.405)

Key: tenths = $\frac{1}{10}$, hundredths = $\frac{1}{100}$, thousandths = $\frac{1}{1000}$

Part A – Name the Place & Value

Number	Underlined digit	Place Name	Value
4 <u>7</u> .305	7		
12.4 <u>8</u> 6	8		
0. <u>0</u> 53	0		
304.0 <u>9</u> 1	9		
5.7 <u>3</u> 8	3		

Part B – Write in Words and Standard Form

1. Write 8.047 in words:

2. Write 0.305 in words:

3. "Fourteen and sixty-two thousandths" =

4. "Three hundred and eight tenths" =

Part C – Decimals as Fractions

5. $0.7 = \frac{\boxed{}}{\boxed{}}$

6. $0.43 = \frac{\boxed{}}{\boxed{}}$

7. $0.125 = \frac{\boxed{}}{\boxed{}}$

8. $\frac{9}{100} = \boxed{}$

Part D – Compare & Order Decimals

Line up the decimal points! Compare digit by digit from left to right.

9. $0.6 \boxed{} 0.60 \boxed{} 0.600$

10. $3.45 \boxed{} 3.450 \boxed{} 3.405$

11. Order from least to greatest:
0.8, 0.08, 0.80, 0.081

12. Order from greatest to least:
4.07, 4.7, 4.071, 4.70

Summer Math Adventure

W4
D2

Week 4 · Day 2 | Rounding Decimals & Decimal Number Lines

Name: **Challenge – Decimal Detectives**

Date: _____

ROUNDING DECIMALS
A decimal is between 0.5 and 0.6. My hundredths digit is 3 more than my tenths digit. What could it be? (find two possibilities)

Rounding decimals — same rule as whole numbers:

Look at the digit ONE place to the RIGHT of where you're rounding.

0–4 → round down, 5–9 → round up.

Round 4.376 to the nearest tenth: look at hundredths digit ($7 \geq 5$) → round up → **4.4**

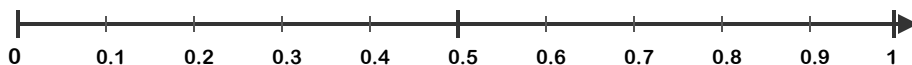
Round 4.376 to the nearest hundredth: look at thousandths digit ($6 \geq 5$) → round up → **4.38**

Part A – Round the Decimals

Number	Nearest whole	Nearest tenth	Nearest hundredth
6.382			
0.547			
14.905			
2.999			
0.0047			

Part B – Number Line: Locate the Decimal

1. Mark and label 0.3, 0.7, and 0.15 on the number line below.



2. What decimal is exactly halfway between 0.4 and 0.5?

3. What decimal is exactly halfway between 3.6 and 3.7?

Part C – Real-World Rounding

4. A bag of chips weighs 8.637 oz. Round to the nearest tenth:

5. Gas in Peoria costs \$3.879 per gallon. A news report rounds to the nearest cent (hundredth). What does the report say?

6. A runner's time is 9.843 seconds. Rounded to the nearest tenth, was it faster or slower than 9.9 seconds?

7. 2,047,300 rounded to nearest million:

8. $\text{GCF}(40, 56) =$

9. $\frac{5}{6} + \frac{1}{4} =$

Summer Math Adventure

W4
D3

Week 4 · Day 3 | Adding & Subtracting Decimals

Name: Challenge – Decimal Round-Trip

Date: _____

A number rounds to 4.6 when rounded to the nearest tenth. What is the largest possible value of the number (to the thousandths place)? What is the smallest?

DECIMAL OPERATIONS

The golden rule: **LINE UP THE DECIMAL POINTS!**

Then add zeros to make the same number of decimal places. Carry and borrow just like whole numbers.

Example: $12.4 + 3.067$

12.400

+ 3.067

15.467 ← decimal point stays in same column!

Part A – Set It Up & Solve

Rewrite vertically with decimal points aligned, add zeros where needed.

1. $4.3 + 2.85 =$

2. $13.7 + 0.048 =$

3. $8.4 - 3.27 =$

4. $10 - 4.375 =$

5. $6.003 + 14.97 + 0.8 =$

6. $25.06 - 18.4 =$

Part B – Error Hunt

7. A student says: $3.4 + 0.06 = 3.46$? Is this right? Circle: YES / NO
If wrong, what did they do wrong?

Correct answer:

8. A student says: $5 - 1.2 = 4.2$? Is this right? Circle: YES / NO
Correct answer:

Part C – Word Problems

9. At the Peoria farmer's market: tomatoes cost \$2.49/lb, apples cost \$1.75/lb, and peppers cost \$3.08/lb. If you buy 1 lb of each, what is the total cost?

Summer Math Adventure

W4
D4

Week 4 · Day 4 | Multiplying Decimals

Name: \$ _____ Date: _____

MULTIPLYING DECIMALS

How to multiply decimals:
Step 1: Ignore the decimal points and multiply as whole numbers.
Step 2: Count total decimal places in BOTH numbers.
Step 3: Place the decimal point that many places from the RIGHT in your answer.
Example: $3.4 \times 0.7 \rightarrow 34 \times 7 = 238 \rightarrow 2$ decimal places total $\rightarrow 2.38$
Sense check: $3.4 \times 0.7 \approx 3 \times 0.7 = 2.1$. Answer 2.38 is close. ✓

Part A – Multiply (Count the decimal places!)

10. Three siblings' heights: Alex is 4.18 feet, Bella is 4.6 feet, Carlos is 5.17 feet. How much taller is Carlos than Alex? What is the combined height of all three?

Whole: $24 \times 13 =$ _____
Decimal places: _____
Answer: _____

Whole: _____
Decimal places: _____
Answer: _____

Whole: _____
Decimal places: _____
Answer: _____

Difference: $0.7 =$ _____
Combined: _____
Answer: _____

$5.314 \times 2.5 =$ _____
Answer: _____

$6.025 \times 0.04 =$ _____
Answer: _____

★ Challenge – Magic Decimal Square

Part B – Decimal x Whole Number
Fill in the blanks: the sum is the same, the product is the same, and diagonal adds to 1.5:

7. $3.6 \times 0.7 =$ _____

		0.3
	0.5	
0.2		0.8

8. $12.4 \times 5 =$ _____

9. $0.045 \times 8 =$ _____

Part C – Powers of 10 Shortcut

Work:
Multiplying by 10, 100, 1000? Move the decimal RIGHT. By 0.1, 0.01? Move it LEFT.

10. $4.7 \times 10 =$ _____

11. $4.7 \times 100 =$ _____

12. $4.7 \times 0.1 =$ _____

13. $4.7 \times 1,000 =$ _____

14. $4.7 \times 0.01 =$ _____

15. $0.006 \times 10 =$ _____

Part D – Word Problems

16. A car gets 32.4 miles per gallon. The gas tank holds 12.5 gallons. How far can the car travel on a full tank?

Summer Math Adventure

W4
D5

Week 4 - Day 5 | Dividing Decimals & Cumulative Review

miles

Name: _____

Date: _____

17. Fabric costs \$4.75 per yard. You buy 3.5 yards. How much do you pay?

DIVIDING DECIMALS

Dividing a decimal by a whole number: Place decimal in quotient directly above the decimal in the dividend. Divide as usual.

$$6.48 \div 4 = 1.62 \quad (\text{think: } 648 \div 4 = 162, \text{ then place decimal: } 1.62)$$

Dividing by a decimal: Multiply BOTH numbers by 10/100 to make divisor a whole number.

$$3.6 \div 0.4 = 36 \div 4 = 9 \quad (\text{multiply both by } 10)$$

★ Challenge – Decimal Patterns

Part A – Decimal ÷ Whole Number

Complete the pattern and explain the rule:

$$1. \quad 1.4007 \div 0.003 = 3 \quad 100 \times 0.003 = 0.3 \quad 14.414 \div 0.003 = \underline{\hspace{2cm}} \quad 1 \times 0.003 = 0.003 \quad 3.796 \div 0.003 = \underline{\hspace{2cm}}$$

Rule:

Use the pattern to find: $0.01 \times 0.003 =$ _____

4. $32.5 \div 5 =$ _____

5. $7.35 \div 3 =$ _____

6. $4.56 \div 12 =$ _____

Part B – Decimal ÷ Decimal

Multiply both numbers to eliminate the decimal in the divisor first.

7. $2.4 \div 0.6 =$ _____

→ $\underline{\hspace{1cm}} \div \underline{\hspace{1cm}} =$ _____

8. $3.5 \div 0.7 =$ _____

→ $\underline{\hspace{1cm}} \div \underline{\hspace{1cm}} =$ _____

9. $0.36 \div 0.04 =$ _____

→ $\underline{\hspace{1cm}} \div \underline{\hspace{1cm}} =$ _____

Part C – Word Problems

10. You and 4 friends split a restaurant bill of \$48.75 equally. How much does each person pay?

Each pays: \$ _____

11. A rope is 13.5 meters long. It is cut into pieces of 0.9 meters each. How many pieces are there?

Answer: _____

Part D – Cumulative Review: Weeks 1–4

12. $5,000,000 - 2,738,416 =$

13. $347 \times 82 =$

Week 4 Quiz – Decimals

W4
QUIZ

Topics: Place Value · Rounding · Add/Subtract · Multiply · Divide

Name: $14.4\frac{1}{3} - 1\frac{5}{6} =$

Date: $15.4.83 + 12.6 + 0.097 =$

Score: / 20

Section 1 – Place Value & Rounding (5 pts)

1. What is the value of 4 in 12.346?

2. Write 0.305 as a fraction.

★ Challenge – Shopping Smarts

At a store:

3. Order: 0.5, 0.05, 0.51, 0.505 (least to greatest)
3 packs of gum: \$0.89 each • 2 water bottles: \$1.45 each • 1 notebook: \$3.99

4. Round 7.4852 to nearest hundredth:
You pay with \$10.00. How much change do you get? If you split the cost 3 ways, how much per person?

5. (1 pt) Write in standard form: "fifty-two and seven hundredths" =

Section 2 – Add & Subtract (5 pts)

6. $8.4 + 3.076 =$

7. $20 - 7.38 =$

Per person: \$

8. (3 pts) Mei ran 3.75 km on Monday, 4.2 km on Wednesday, and 2.875 km on Friday. How many total km did she run? Round your answer to the nearest tenth.

Total:

km Rounded:

km

Section 3 – Multiply & Divide (10 pts)

9. $2.4 \times 3.5 =$

10. $0.07 \times 0.9 =$

11. $6.48 \div 4 =$

12. $4.5 \div 0.9 =$

13. (2 pts) A bag of trail mix weighs 1.35 kg. There are 6 bags. What is the total weight?

Answer:

kg

14. (4 pts) You earn \$12.50 per hour. You work 8 hours on Saturday and 6.5 hours on Sunday. After spending \$24.75 on lunch both days, how much money do you have left?

Answer: \$
